MIL-G-8602A
6 APRIL 1956
SUPERSEDING
MIL-G-8602
29 JUNE 1953

MILITARY SPECIFICATION

GLASS, LAMINATED, FLAT, AIRCRAFT

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 Scope. This specification covers laminated flat glass suitable for use in aircraft glazing.

1.2 Classification. Laminated flat glass shall be of the following classes, types, and grades, as specified. (See 6.2.)

Class I—Block, annealed glass
- Type A—Relaxed optical requirements 1 inch from edge of glass

Class II—Ready-cut, annealed, or tempered, as specified
- Type A—Relaxed optical requirements up to 2 inches from the edge of the daylight opening.
- Type B—Relaxed optical requirements up to 1 inch from the edge of the daylight opening.
- Type C—Unrelaxed optical requirements over entire daylight opening.

Grade 2N—2-minute deviation
Grade 3N—3-minutes normal deviation
Grade 4.5N—4.5-minutes normal deviation
Grade 7N—7-minutes normal deviation

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, and publications, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

DD-G-451	—Glass, Flat and Corrugated, for Glazing, Mirrors, and Other Uses
NN-B-591	—Boxes, Fiberboard, Wood-Cleated (For Domestic Shipment)
NN-B-621	—Boxes, Wood, Nailed and Lock-Corner
NN-B-631	—Boxes, Wood, Wirebound, (For Domestic Shipment)
NN-P-515	—Plywood, Container Grade
UU-P-268	—Paper, Kraft, Wrapping
PPP-B-601	—Boxes, Wood, Cleated-Plywood

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JAN-P-100	—Packaging and Packing for Overseas Shipment; General Specification
JAN-P-103	—Packaging and Packing for Overseas Shipment—Boxes; Wood Cleated; Solid Fiberboard
JAN-B-106	—Boxes, Wood, Nailed (Overseas Type)

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MIL-B-107 —Boxes, Wood, Wirebound (Military Type)

MIL-P-116 —Preservation, Methods of

MIL-B-138 —Boxes, Wood, Fiberboard-Lined for Overseas Shipment (For Weight Contents not Exceeding 500 Pounds)

MIL-B-10377 —Box, Wood, Cleated, Veneer, Paper Overlaid

MIL-L-10547 —Liners, Case, Waterproof

STANDARDS

MILITARY

MIL-STD-129—Marking for Shipment and Storage

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other Publications. The following document forms a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

American Society for Testing Materials Standards

D-1003-52 Haze and Luminous Transmittance of Transparent Plastics

(Copies of A.S.T.M. specifications may be obtained from the American Society for Testing Materials, 1916 Race Street, Philadelphia 3, Pennsylvania.)

3. REQUIREMENTS

3.1 Materials

3.1.1 *Glass.*—The glass plies used in the construction of flat laminated panels shall be specially selected, glazing quality, polished plate glass conforming to Specification DD-G-451, type I, except as to thickness. All glass used in class I lamination shall be in the annealed condition. Class II glass may be annealed or tempered as specified by the procuring activity.

3.1.2 *Plastic.* The plastic used in the interlayer shall be stable without edge sealing and shall have suitable binding characteristics. The plastic shall be either polyvinyl butyral plasticized with 21.0 (plus or minus 2) parts by weight of triethylene glycol di-2-ethyl butyrate (3GH) per 100 parts resin or a silicone-type material.

3.2 Design and Construction. Laminated glass shall consist of two or more pieces of glass held together by interlayers of transparent plastic having suitable binding characteristics. Class II glass may include extended plastic edges with or without metal reinforcements as specified on the applicable drawing and contract or order. When specified in those class II assemblies in which components are of dissimilar size, the plastic which extends beyond the edge of the smaller piece of glass shall not be bonded to the extended area of the larger piece of glass.

3.3 Glass Quality Uniformity. The finished glass assembly shall contain no defects in excess of the additive total of those permitted in the individual components. The assembly shall be suitable for viewing objects at 20 feet or more from the panel when held at not more than 3 feet from the observer.

3.3.1 *Unobstructed Vision Area.* The unobstructed vision area is that area of the glass panel, the visibility of which is not obstructed by other components of the airplane such as the nose. This area shall contain no defects in excess of the additive total of those permitted by Specification DD-G-451, type I, in the individual plies before lamination. Occasional minute lint particles, shell cullet, and light plastic streaks are permitted in the plastic interlayers, provided these defects do not impair visibility. Only occasional very light scratches, not more than 3 inches long which do not impair visibility, are permitted on the surface.

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3.3.2 *Obstructed Vision Area.* The obstructed vision area is that area of a glass panel through which visibility is obstructed by other components of the airplane such as the nose. This area may contain the above defects and, in addition, may contain an occasional piece of bristle, light finger prints, and very light scratches, provided these defects do not impair visibility. The area of the glass covered by the mounting frame may contain any defects, including blow-ins 1 by ¼ inches along the edge, which do not impair the strength of the glass and which do not cause objectionable reflections or scatter of light. Vents and deep-edge chips are not permissible.

3.4 Resistance to Temperature Change. The glass assembly shall withstand the temperature change test (see 4.3.6) without cracking, clouding, delamination, formation of bubbles, or other evidence of deterioration. For class II glass containing components of dissimilar size, the separation of small flakes or chips beyond the trim line of the smaller glass shall not be considered evidence of deterioration.

3.5 Original Luminous Transmittance. The original luminous transmittance shall be at least 80 percent for assemblies of overall thickness up to ⅝ inch. The specified transmittance of assemblies greater than ⅝ inch shall be at least 72 percent.

3.6 Luminous Transmittance after Exposure. After 100 hours exposure to the light exposure test specified in section 4, the luminous transmittance shall conform to the requirement for original luminous transmittance.

3.7 Optical Deviation. The optical deviation of parallel light at normal incidence to the glass shall not exceed the values shown in table I when the glass is subjected to the optical deviation test specified in section 4. Unless otherwise specified, relaxed optical deviation is permissible in the margin 1 inch from the edge of class I glass and in the margin 2, 1, and 0 inches from the edge of the daylight opening of class II assemblies, depending on type.

TABLE I

Normal incidence deviation data

Grade	Optical deviation (minutes of arc)	Projection image shift at 25 ft (maximum in inches from undeviated position)
2N	2	0.18
3N	3	0.26
4.5N	4.5	0.38
7N	7	0.56

3.8 Nonscatterability. Glass subjected to the impact test shall show no separation of glass and interlayer except for a ¼-inch diameter spot at point of impact. Very small amounts of glass may leave the underside of the assembly because of fracture of the bottom plate.

3.9 Finish. Class I glass shall have clean-cut edges. This requirement permits class I glass to have unground areas or "shiners" on the edges. Class II glass which is tempered will be allowed a maximum slippage tolerance of 1/16 inch. Other class II glass shall have exposed edges fully ground flush with no overlapping plates and no sharp edges unless otherwise specified by the applicable drawing and contract or order.

3.10 Tolerance.

3.10.1 Unless otherwise specified by the procuring activity, the following thickness tolerances will be allowed. Thickness tolerance for individual plies shall be as indicated in Specification DD–C–451. The assembly

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shall have a thickness tolerance of plus or minus 1/32 inch for assemblies of 3/8 inch or below and plus or minus 1/8 inch for other assemblies.

3.10.2 *Linear Dimensions.* Unless otherwise specified by the procuring activity the following linear dimensional tolerances will be allowed. Annealed glass assemblies 1/4 inch or under in thickness, class I or II, will be allowed a linear dimensional tolerance of plus or minus 1/32 inch. Annealed glass assemblies over 1/4 inch thick, class I or II, will be allowed a linear dimensional tolerance of plus or minus 1/16 inch. Tempered glass assemblies regardless of thickness will be allowed a linear dimensional tolerance of plus or minus 1/16 inch.

3.10.3 *Portion of Metal Reinforcement.* That portion of the metal reinforcement underlying the glass shall be held at a minimum consistent with good commercial practice

3.11 **Identification of Product.** Each glass assembly shall be indelibly marked according to the following form by etching or sandblasting on the glass surface. Unless otherwise specified, the marking shall not extend more than 5/8 inch from the nearest glass edge and shall not be longer than 2 inches. This requirement shall apply only to class II glass.

Identification form:

Drawing Number Date (Month-Year)	*Specification Number Manufacturer's Trademark*	*Class, Type and Grade Manufacturer's Serial Number*
Example:		
22-1067-1 10-55	8602 XYZ	11A2N 774210

3.12 **Workmanship.** Workmanship shall be in accordance with the best practice for high-grade aircraft laminated glass.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Classification of Tests.** The inspection and testing of the laminated flat glass shall be classified as follows:

Acceptance tests

4.2 **Acceptance Tests.** Acceptance tests shall consist of:

a. Individual tests
b. Sampling tests

4.2.1 *Individual Tests.* Each assembly and laminated sheet shall be subjected to the following tests as described under 4.3 "Test Methods."

a. Examination of product
b. Optical deviation

4.2.2 Sampling Plans and Tests.

4.2.2.1 *Lot.* A lot shall consist of all glass assemblies prepared by the same process, of the same nominal thickness (disregarding over-all rectilinear dimensions), the same class, type, and grade, and laminated within any 30-day period. (A limited shipment for experimental airplanes shall not be considered a lot.)

4.2.2.2 *Sampling Plan.* Unless otherwise specified, 2 glass assemblies form each 1000 assemblies, or less, in each lot shall be selected at random and subjected to the following tests in the order named:

Sample No. 1	Sample No. 2
a. Original luminance transmittance	Resistance to temperature change
b. Luminous transmittance after exposure	
c. Nonscatterability	

4.2.2.2.1 Production glass assemblies rejected for other reasons may be used for these tests

4.2.3 *Rejection and Retest.* Failure of any specimen in any requirement for which it represents a lot shall cause rejection of the lot represented. Failure of any assembly in any requirement for which all assemblies are inspected shall cause rejection of the individual assembly. Rejected glass assemblies shall not be resubmitted for test without full particulars concerning previous rejection being given the inspector.

4.3 **Test Methods.**

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4.3.1 *Examination of Product.* Each assembly and laminated sheet shall be carefully examined to determine conformance to the requirements of this specification with respect to dimensions, workmanship, glass quality uniformity, finish and identification of product.

4.3.1.1 Each transparent assembly shall be visually inspected for the requirements of 3.3 in the equivalent of light from a clear sky without sun when placed in a position similar to that of a glazed light with the examiner's eye on a level with the center of the assembly and looking through the assembly from a distance of about 36 inches against a brick wall or similar background, showing essentially straight lines.

4.3.2 *Optical Deviation.* The optical deviation shall be determined by examining the image projected on a screen through the specimen during a complete survey of its area or by the alternative method in 4.3.2.1 under the conditions stated. The projection (line) method shall be considered the reference method in case of a disagreement. A lantern-slide projector shall be used to project the image of a narrow black line focused on a smooth screen. The screen shall be located in a fixed position at a distance of 15 to 25 feet from the specimen. An opaque shield having a vertical aperture approximately 1 by 10 inches shall be mounted 3 feet in front of the projector lens. The screen shall contain a dashed vertical line and 8 paralled black lines, 4 on either side, corresponding to angular deviations of 2, 3, 4.5, and 7 minutes, respectively, from the center line. Suitable dimensions for a specimen-to-screen distance of 25 feet are given in table I. The screen shall be alined with the projector to provide superposition of the central dashed line and the projected line. The specimen shall be passed through the light beam in its own plane normal to the beam. After the specimen has been moved horizontally until each portion of the area being tested has intercepted the beam, it shall be rotated through 90° in its own plane and the test repeated. In both positions the deviation on the screen shall be continuously noted. Any deviation of any part of the projected vertical line from the central dashed line exceeding the amount specified for the grade required shall be cause for rejection. Observations may be made by means of a second observer at the screen or by means of a low-power telescope.

4.3.2.1 *Alternative Optical Deviation.* The optical deviation may be determined as specified above by observing the position of the secondary image, caused by lack of-parallelism between the two surfaces, relative to the primary image of a small luminous dot. The light box shown in figure 1 shall be viewed against a black background having a maximum illumination of 8 foot-candles. The face of the light box shall be black with the four white or translucent circles and a center aperture covered by a red filter. The specimen shall be from 15 to 25 feet from the light box and shall be so positioned that the area being examined will be in or tangent to a plane normal to the line of sight between the examiner's eye and the light source. The light box circles shall have the dimensions appropriate to the specimen to light box distance used. The examiner's eye shall be within 3 feet of the specimen. The specimen shall be systematically moved across the fixed line of sight in order that the examination includes every portion of the entire area except the specified margin around the edge of the glass. Scanning shall be done by moving the specimen in each of 2 directions approximately 90° apart. The observer shall use only one eye at a time. The location of the observer's eye shall be on a line normal to the face of the light box passing through the central aperture. The displacement of the secondary image outside of the circle of the radius corresponding to the specified angular deviation shall be cause for rejection.

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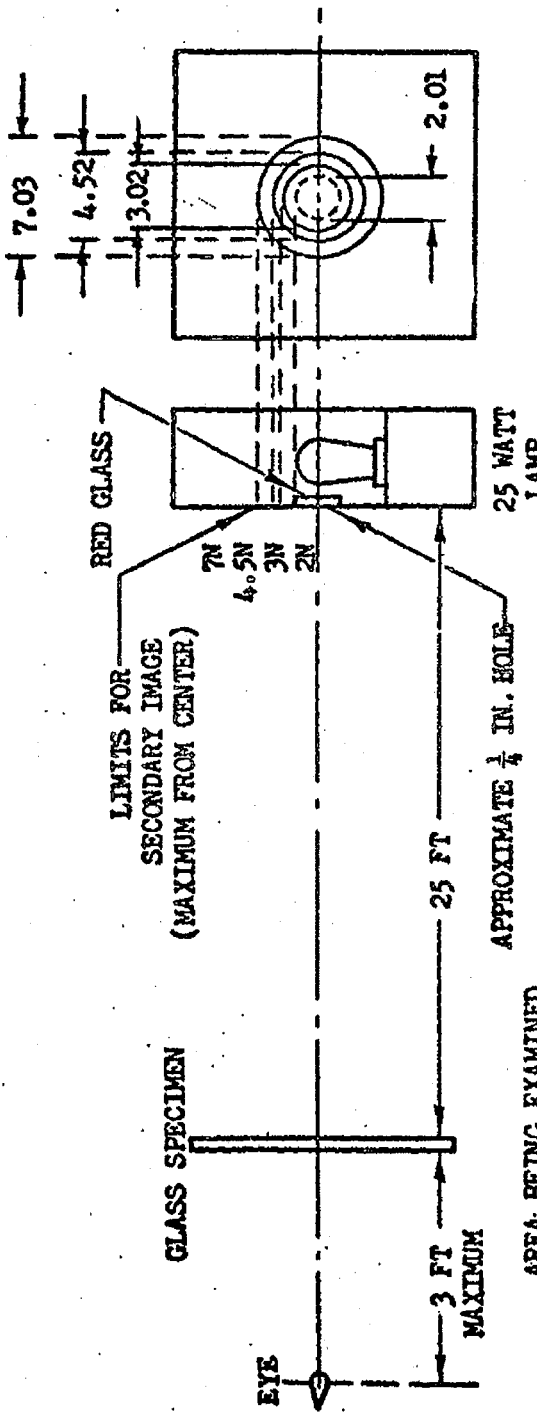


FIGURE 1. *Optical deviation test.*

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4.3.3 *Original Luminous Transmittance.* The luminous transmittance of the sample shall be determined in accordance with the test for luminous transmittance of transparent plastic, using illuminant C, specified in ASTM Designation D 1033-52 or equivalent method.

4.3.4 *Luminous Transmittance after Exposure.* The luminous transmittance of the specimen after exposure shall be determined in the same manner as for original luminous transmittance. The light source used in the exposure of the specimen shall be a flaming carbon arc of the Fade-O-Meter type inclosed in a clear Corax D globe, using the equivalent of Atlas No. 70 and No. 220 carbons, with a current of 13 amperes or a quartz mercury arc possessing controls to provide an intensity equivalent to the flaming carbon arc. The specimen shall be exposed at a distance of 15 inches from the arc with the plane of the glass normal to the direction of the source. The exposure time shall be 100 hours. The specimen temperature shall not exceed +100°C (+212°F) at any time during the exposure period.

4.3.5 *Nonscatterability.* The test specimens shall be subjected to impact when supported in a horizontal position by a suitable frame. This frame shall support the entire length of each side of the glass along a margin not more than ⅜ inch wide. The frame shall rest on a metal plate of sufficient weight and rigidity to prevent distortion. A ½-pound spherical steel ball dropped from a height of 16 feet with no initial velocity or force applied shall strike the approximate center of the specimen with a single gravity-accelerated blow. In case all glass plies are not fractured by this impact, the specimen shall be turned upside down and the falling-ball impact repeated. If all glass plies are still not fractured, the glass assembly shall be subjected to impact by the minimum number of hammer strokes required to cause fracture of all glass plies. Immediately after all glass plies have been fractured, the assembly shall be examined to detect any interlayer surface which is free from glass. Such an area is considered evidence of separation of glass and the interlayer. The specimens shall be tested at a temperature +23° plus or minus 1.1°C (+73.5° plus or minus 2°F).

4.3.6 *Resistance to Temperature Change.* Specimens shall be placed in air at a temperature of −54° plus or minus 5°C (−65° plus or minus 10°F) for a period of 2 hours after temperature equilibrium has been attained in the assembly, then placed in still air at a temperature of +23° plus or minus 1.1°C (+73.5° plus or minus 2°F) for 1 hour and then placed in circulating air removal to still air at a temperature of +23° plus or minus 1.1°C (+73.5° plus or minus 2°F) and cooling to this temperature, the specimen shall be examined for evidence of deterioration caused by the test.

5. PREPARATION FOR DELIVERY

5.1 Levels of Preservation and Packaging.

5.1.1 *Level A and Level B.* Military Package and Limited Military Package (For Overseas Shipment and Domestic Shipment with more than 180 Days Storage).—Glass shall be preserved in accordance with Specification MIL-P-116, method III. Each individual assembly of laminated glass shall have its surfaces covered with newsprint and shall be wrapped in 30 pound or heavier kraft paper conforming to Specification UU-P-268, grade B. Unit quantities shall be as specified by the procuring agency.

5.1.2 *Level C.* Minimum Military Package (For Domestic Shipment with Immediate Use at Initial Destination).—Glass shall be preserved and packaged in accordance with manufacturer's commercial practice.

5.2 Levels of Packing.

5.2.1 *Level A.* Military Pack (Overseas Shipment).—Glass preserved and packaged to meet 5.1.2 shall be packed in exterior type shipping containers meeting Specifications JAN-P-103, JAN-P-106, MIL-B-138, MIL-B-107, or PPP-B-601 (Overseas Type), respectively. Plywood shall be type I or II,

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class 2 conforming to Specification NN-P-515. As far as practical, exterior containers shall be of uniform shape and size, be of minimum cube and tare consistent with the protection required, and contain identical quantities. The gross weight of each pack shall be limited to approximately 200 pounds. Containers shall be closed and strapped in accordance with the applicable container specification or appendix thereto. Containers shall be provided with a case liner conforming to Specification MIL-L-10547 and shall be sealed in accordance with the appendix thereto,

5.2.2 *Level B.* Limited Military Pack (For Domestic Shipment and Storage).—Glass preserved and packaged to meet 5.1.2 shall be packed in domestic type exterior containers meeting Specification NN-B-591, PPP-B-601 (Domestic Type), NN-B-621, NN-B-631, or MIL-B-10377 (Domestic Type), respectively. Exterior containers shall be of minimum cube and tare consistent with the protection required. As far as practical, exterior containers shall be of uniform shape and size and contain identical quantities. The gross weight of each pack shall be limited to approximately 500 pounds. Containers shall be closed and strapped in accordance with the applicable container specification or appendix thereto.

5.2.3 *Level C.* Minimum Military Pack (For Domestic Shipment with Immediate use at Initial Destination).—Packages which require overpacking for acceptance by the carrier shall be packed in exterior type shipping containers in a manner that will insure safe transportation at the lowest rate to the point of delivery. Containers shall meet Consolidated Freight Classification Rules or regulations of other common carriers, as applicable to the mode of transportation.

5.3 **Physical Protection.** Cushioning, blocking, bracing, and bolting as required shall be in accordance with Specification JAN-P-100 except that for domestic shipments waterproofing requirements for cushioning materials and containers shall be waived. Drop tests of Specification JAN-P-100 shall be waived when preservation, packaging, and packing of the item is for immediate use or when drop tests of Specification MIL-P-116 are applicable.

5.4 **Marking of Shipments.** Interior packages and exterior shipping containers shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Intended Use

6.1.1 *Class I.* Class I glass is intended for general glazing purposes in which rectangular shapes may be used without cutting or irregular shapes may be cut at the time and place of installation. Grade 2N glass is intended for glazing use where undistorted vision is especially desirable to insure minimum error in navigation and other accurate observations. This grade represents a selected portion of production and is of limited availability, particularly for larger size assemblies. Grade 3N glass is intended for use in most aircraft glazing where vision is required. Grade 4.5N and 7N is intended for use on these assemblies where very thick interlayers are used.

6.1.2 *Class II.* Class II glass is intended for glazing installations for specific aircraft. Grade 2N glass is intended for glazing use where undistorted vision is especially desirable to insure minimum error in navigation and other aircraft observation. Grade 3N glass is intended for use in most aircraft glazing where vision is required. Grade 4.5N and 7N is intended for use on these assemblies where very thick interlayers are used.

6.2 Ordering Data

6.2.1 *Class I.* Requisitions, contracts, and orders should specify class, type and grade, size and thickness, the quantity desired, and whether overseas packing should be furnished. In general, it should be noted that the yield of large glass assemblies of a given quality is less than for small glass assemblies and that cost per unit area increases with

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size. Large size assemblies should therefore not be procured for recutting into small blocks.

6.2.2 *Class II.* Requisitions. contracts, and orders should specify class, type and grade, part number, the quantity desired, and whether overseas packing should be furnished.

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Preparing Activity:
Air Force

Custodians:
Army—Transportation Corps
Navy—Bureau of Aeronautics
Air Force

Other interest:
Army—CQ
Navy—OrS

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